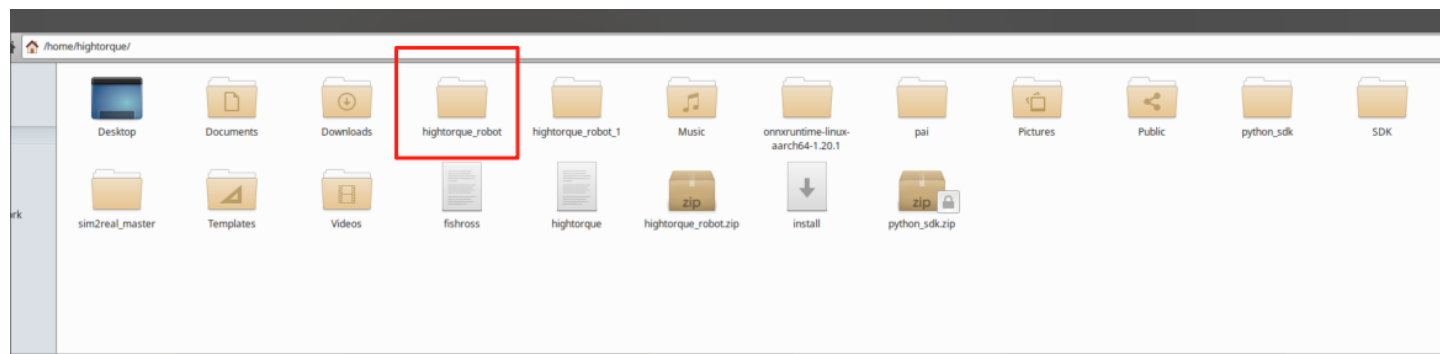


01、python-SDK环境配置

python-SDK环境配置

1. 下载程序

将程序下载在主机中，文件名为hightorque_robot



2. 下载系统依赖

- 打开终端输入以下指令

代码块

```
1  sudo apt-get update
2  sudo apt-get install -y \
3      cmake \
4      build-essential \
5      libserialport-dev \
6      libyaml-cpp-dev
```

- 运行结果如图所示

```
Terminal - hightorque@lubancat: ~/hightorque_robot
File Edit View Terminal Tabs Help
hightorque@lubancat:~/hightorque_robot$ sudo apt-get update
Hit:1 http://mirrors.tuna.tsinghua.edu.cn/ros/ubuntu focal InRelease
Get:2 http://mirrors.tuna.tsinghua.edu.cn/ros2/ubuntu focal InRelease [4685 B]
Hit:3 https://packages.microsoft.com/repos/code stable InRelease
Hit:4 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal InRelease
Hit:5 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates InRelease
Hit:6 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-backports InRelease
Hit:7 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-security InRelease
Fetched 4685 B in 3s (1710 B/s)
Reading package lists... Done
hightorque@lubancat:~/hightorque_robot$ sudo apt-get install -y \
> cmake \
> build-essential \
> libserialport-dev \
> libyaml-cpp-dev
```

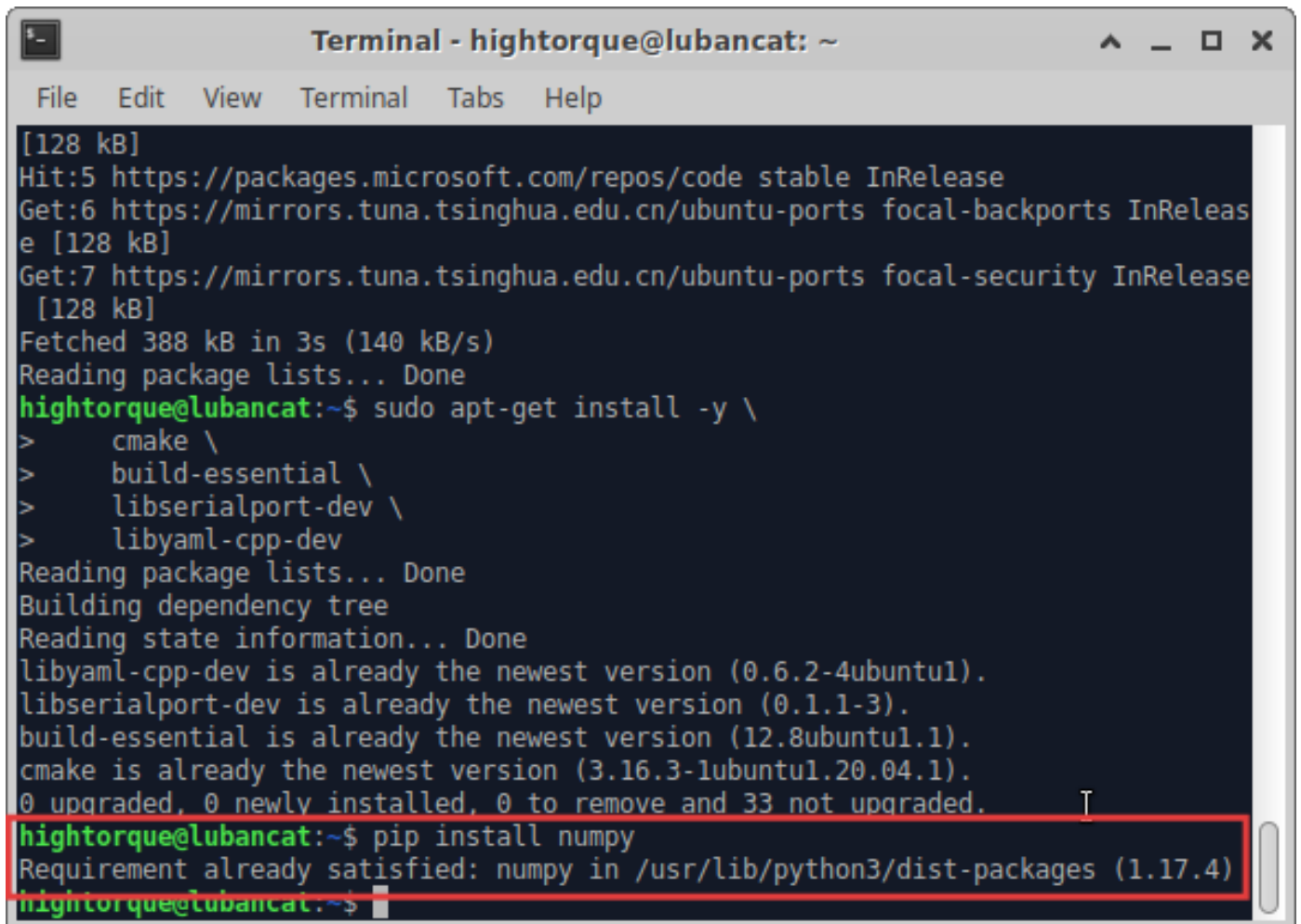
```
Terminal - hightorque@lubancat: ~/hightorque_robot
File Edit View Terminal Tabs Help
Hit:1 http://mirrors.tuna.tsinghua.edu.cn/ros/ubuntu focal InRelease
Get:2 http://mirrors.tuna.tsinghua.edu.cn/ros2/ubuntu focal InRelease [4685 B]
Hit:3 https://packages.microsoft.com/repos/code stable InRelease
Hit:4 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal InRelease
Hit:5 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates InRelease
Hit:6 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-backports InRelease
Hit:7 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-security InRelease
Fetched 4685 B in 3s (1710 B/s)
Reading package lists... Done
hightorque@lubancat:~/hightorque_robot$ sudo apt-get install -y \
> cmake \
> build-essential \
> libserialport-dev \
> libyaml-cpp-dev
Reading package lists... Done
Building dependency tree
Reading state information... Done
libyaml-cpp-dev is already the newest version (0.6.2-4ubuntu1).
libserialport-dev is already the newest version (0.1.1-3).
build-essential is already the newest version (12.8ubuntu1.1).
cmake is already the newest version (3.16.3-1ubuntu1.20.04.1).
0 upgraded, 0 newly installed, 0 to remove and 33 not upgraded.
hightorque@lubancat:~/hightorque_robot$
```

- 安装Python绑定依赖，运行下列指令

代码块

```
1 pip install numpy
```

- 运行结果如下图



A terminal window titled "Terminal - hightorque@lubancat: ~" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal output shows the installation of various packages using apt-get. The packages listed are cmake, build-essential, libserialport-dev, and libyaml-cpp-dev. The output indicates that these packages are already the newest versions. The final command executed is `pip install numpy`, which results in the message "Requirement already satisfied: numpy in /usr/lib/python3/dist-packages (1.17.4)". The last line of the terminal output is `hightorque@lubancat:~$`.

```
[128 kB]
Hit:5 https://packages.microsoft.com/repos/code stable InRelease
Get:6 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-backports InRelease [128 kB]
Get:7 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-security InRelease [128 kB]
Fetched 388 kB in 3s (140 kB/s)
Reading package lists... Done
hightorque@lubancat:~$ sudo apt-get install -y \
> cmake \
> build-essential \
> libserialport-dev \
> libyaml-cpp-dev
Reading package lists... Done
Building dependency tree
Reading state information... Done
libyaml-cpp-dev is already the newest version (0.6.2-4ubuntu1).
libserialport-dev is already the newest version (0.1.1-3).
build-essential is already the newest version (12.8ubuntu1.1).
cmake is already the newest version (3.16.3-1ubuntu1.20.04.1).
0 upgraded, 0 newly installed, 0 to remove and 33 not upgraded.
hightorque@lubancat:~$ pip install numpy
Requirement already satisfied: numpy in /usr/lib/python3/dist-packages (1.17.4)
hightorque@lubancat:~$
```

3. 编译程序

- 进入hightorque_robot文件并建立build文件。
 - 输入以下指令

代码块

```
1 cd hightorque_robot
2 mkdir build && cd build
```

- 运行结果如下图

```
Terminal - hightorque@lubancat: ~/hightorque_robot/build
File Edit View Terminal Tabs Help
Get:3 http://mirrors.tuna.tsinghua.edu.cn/ros2/ubuntu focal InRelease [4685 B]
Hit:4 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal InRelease
Hit:5 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates InRelease
Hit:6 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-backports InRelease
Hit:7 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-security InRelease
Fetched 4685 B in 3s (1483 B/s)
Reading package lists... Done
hightorque@lubancat:~$ sudo apt-get install -y \
>     cmake \
>     build-essential \
>     libserialport-dev \
>     libyaml-cpp-dev
Reading package lists... Done
Building dependency tree
Reading state information... Done
libyaml-cpp-dev is already the newest version (0.6.2-4ubuntu1).
libserialport-dev is already the newest version (0.1.1-3).
build-essential is already the newest version (12.8ubuntu1.1).
cmake is already the newest version (3.16.3-1ubuntu1.20.04.1).
0 upgraded, 0 newly installed, 0 to remove and 33 not upgraded.
hightorque@lubancat:~$ cd hightorque_robot
hightorque@lubancat:~/hightorque_robot$ mkdir build && cd build
hightorque@lubancat:~/hightorque_robot/build$
```

- 运行编译指令

代码块

```
1  cmake ..
```

- 运行结果如下，确保运行没有报错再执行后面操作。

```
Terminal - hightorque@lubancat: ~/hightorque_robot/build
File Edit View Terminal Tabs Help
hightorque@lubancat:~$ sudo apt-get update
Hit:1 https://packages.microsoft.com/repos/code stable InRelease
Hit:2 http://mirrors.tuna.tsinghua.edu.cn/ros/ubuntu focal InRelease
Hit:3 http://mirrors.tuna.tsinghua.edu.cn/ros2/ubuntu focal InRelease [4645 B]
Hit:4 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal InRelease
Hit:5 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-updates InRelease
Hit:6 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-backports InRelease
Hit:7 https://mirrors.tuna.tsinghua.edu.cn/ubuntu-ports focal-security InRelease
Fetched 4645 B in 3s (1483 B/s)
Reading package lists... Done
hightorque@lubancat:~$ sudo apt-get install -y \
> cmake \
> build-essential \
> libserialport-dev \
> libyaml-cpp-dev
Reading package lists... Done
Building dependency tree
Reading state information... Done
libyaml-cpp-dev is already the newest version (0.6.2-4ubuntu1).
libserialport-dev is already the newest version (0.1.1-3).
build-essential is already the newest version (12.4ubuntu1).
cmake is already the newest version (3.16.3-1ubuntu1.20.04.1).
0 upgraded, 0 newly installed, 0 to remove and 33 not upgraded.
hightorque@lubancat:~$ cd hightorque_robot/
hightorque@lubancat:~/hightorque_robot$ make build --no-build-id
-- The C compiler identification is GNU 9.4.0
-- The CXX compiler identification is GNU 9.4.0
-- Check for working C compiler: /usr/bin/cc -- works
-- Detecting C compiler ABI info -- done
-- Detecting C compiler ABI info - done
-- Detecting C compile features -- done
-- Detecting CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info -- done
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features -- done
-- Setting build type to 'Release' as none was specified.
-- Found PkgConfig: /usr/bin/pkg-config (found version "0.29.1")
-- Checking for module 'glib-2.0'
-- Found glib-2.0, version 2.64.6
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_std_gnu99 - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_std_gnu99 - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wall - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wuninitialized - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wno_format_zero_length - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wno_stringop_truncation - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wall - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wuninitialized - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wno_format_zero_length - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wno_stringop_truncation - Success
-- Building LCM 1.5.1 (ABI v1)
-- Looking for pthread.h - found
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Failed
-- Looking for pthread_create in pthreads - not found
-- Looking for pthread_create in pthread - found
-- Found Threads: TRUE
-- Performing Test COMPILER_HAS_HIDDEN_VISIBILITY - Success
-- Performing Test COMPILER_HAS_HIDDEN_INLINE_VISIBILITY - Success
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
CMake Warning at lib/lcm/docs/CMakeLists.txt:5 (message):
  Doxygen was not found; documentation generation will be incomplete

-- Could NOT find Java (missing: Java_JAVA_EXECUTABLE Java_JAR_EXECUTABLE Java_JAVAC_EXECUTABLE Java_JAVAH_EXECUTABLE Java_JAVADOC_EXECUTABLE) (Required is at least version "1.8")
-- Could NOT find Lua (missing: LUA_LIBRARIES LUA_INCLUDE_DIR)
Unable to parse the Go version string:
-- Could NOT find Go (missing: GO_VERSION GO_PLATFORM GO_ARCH)
-- CPack: Packages will be placed under /home/hightorque/hightorque_robot/build/packages
-- pybind11 v2.11.0 dev1
-- Found PythonInterp: /usr/bin/python3 (found suitable version "3.8.10", minimum required is "3.6")
-- Found PythonLibs: /usr/lib/aarch64-linux-gnu/libpython3.8.so
-- Performing Test HAS_FLTO - Success
-- Performing Test HAS_FLTO - Success
-- Found pybind11:
-- Configuring done
-- Generating done
-- Build files have been written to: /home/hightorque/hightorque_robot/build
hightorque@lubancat:~/hightorque_robot/build$
```

```
Terminal - hightorque@lubancat: ~/hightorque_robot/build
File Edit View Terminal Tabs Help
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wall - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wuninitialized - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wuninitialized - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wno_format_zero_length - Success
-- Performing Test CMAKE_C_COMPILER_SUPPORTS_Wno_stringop_truncation - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wall - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wshadow - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wuninitialized - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wuninitialized - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_variable - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Werror_return_type - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wunused_parameter - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wno_format_zero_length - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wno_stringop_truncation - Success
-- Performing Test CMAKE_CXX_COMPILER_SUPPORTS_Wno_stringop_truncation - Success
-- Building LCM 1.5.1 (ABI v1)
-- Looking for pthread.h - found
-- Performing Test CMAKE_HAVE_LIBC_PTHREAD - Failed
-- Looking for pthread_create in pthreads - not found
-- Looking for pthread_create in pthread - found
-- Found Threads: TRUE
-- Performing Test COMPILER_HAS_HIDDEN_VISIBILITY - Success
-- Performing Test COMPILER_HAS_HIDDEN_INLINE_VISIBILITY - Success
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
CMake Warning at lib/lcm/docs/CMakeLists.txt:5 (message):
  Doxygen was not found; documentation generation will be incomplete

-- Could NOT find Java (missing: Java_JAVA_EXECUTABLE Java_JAR_EXECUTABLE Java_JAVAC_EXECUTABLE Java_JAVAH_EXECUTABLE Java_JAVADOC_EXECUTABLE) (Required is at least version "1.8")
-- Could NOT find Lua (missing: LUA_LIBRARIES LUA_INCLUDE_DIR)
Unable to parse the Go version string:
-- Could NOT find Go (missing: GO_VERSION GO_PLATFORM GO_ARCH)
-- CPack: Packages will be placed under /home/hightorque/hightorque_robot/build/packages
-- pybind11 v2.11.0 dev1
-- Found PythonInterp: /usr/bin/python3 (found suitable version "3.8.10", minimum required is "3.6")
-- Found PythonLibs: /usr/lib/aarch64-linux-gnu/libpython3.8.so
-- Performing Test HAS_FLTO - Success
-- Performing Test HAS_FLTO - Success
-- Found pybind11:
-- Configuring done
-- Generating done
-- Build files have been written to: /home/hightorque/hightorque_robot/build
hightorque@lubancat:~/hightorque_robot/build$
```

运行make指令

代码块

1 make

- 运行结果如下，确保运行没有报错再执行后面操作。

```
Terminal - hightorque@lubancat: ~/hightorque_robot/build
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
-- Performing Test COMPILER_HAS_DEPRECATED_ATTR - Success
-- Make Warning at lib/cn/docs/MakeLists.txt:5 (message):
  Doxygen was not found; documentation generation will be incomplete

-- Could NOT find Java (missing: Java_JAVA_EXECUTABLE Java_JAR_EXECUTABLE Java_JAVAC_EXECUTABLE Java_JAVAH_EXECUTABLE Java_JAVADOC_EXECUTABLE) (Required is at least version "1.8")
-- Could NOT find Lua (missing: LUA_LIBRARIES LUA_INCLUDE_DIRS)
-- Unable to parse the Go version string:
-- Could NOT find Go (missing: GO_VERSION GO_PLATFORM GO_ARCH)
-- Check Packages will be placed under /home/hightorque/hightorque_robot/build/packages
-- pybind11 v2.11.0 dev1
-- Found PythonInterp: /usr/bin/python3 (found suitable version "3.8.10", minimum required is "3.6")
-- Found PythonLibs: /usr/lib/aarch64-linux-gnu/libpython3.8.so
-- Performing Test HAS_FLTO - Success
-- Performing Test HAS_FLTO - Success
-- Found pybind11:
-- Configuring done
-- Generating done
-- Build files have been written to: /home/hightorque/hightorque_robot/build
hightorque@lubancat:~/hightorque_robot/build$ make
Scanning dependencies of target lcn
[ 1%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn.c.o
[ 1%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn_files.c.o
[ 2%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn_memq.c.o
[ 3%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn_mqdpn.c.o
[ 3%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn_topu.c.o
[ 4%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn_udp.c.o
[ 4%] Building C object lib/cn/cn/MakeFiles/lcn.dir/ringbuffer.c.o
[ 5%] Building C object lib/cn/cn/MakeFiles/lcn.dir/udpn_util.c.o
[ 5%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn/types/channel_port_map_update.t.c.o
[ 6%] Building C object lib/cn/cn/MakeFiles/lcn.dir/lcn/types/channel_to_port.t.c.o
[ 6%] Linking C shared library liblcn.so
[ 6%] Built target lcn
Scanning dependencies of target serial
[ 6%] Building CXX object lib/serial/MakeFiles/serial.dir/src/serial.cc.o
[ 7%] Building CXX object lib/serial/MakeFiles/serial.dir/src/impl/unix.cc.o
[ 7%] Building CXX object lib/serial/MakeFiles/serial.dir/src/impl/list_ports/linux.cc.o
[ 8%] Linking CXX shared library libserial.so
[ 8%] Built target serial
Scanning dependencies of target hightorque_robot
[ 8%] Building CXX object MakeFiles/hightorque_robot.dir/src/serial_driver.cpp.o
[ 8%] Building CXX object MakeFiles/hightorque_robot.dir/src/parse_robot_params.cpp.o
[ 9%] Building CXX object MakeFiles/hightorque_robot.dir/src/hardware/motor.cpp.o
[ 10%] Building CXX object MakeFiles/hightorque_robot.dir/src/hardware/canport.cpp.o
[ 10%] Building CXX object MakeFiles/hightorque_robot.dir/src/hardware/canboard.cpp.o
[ 11%] Building CXX object MakeFiles/hightorque_robot.dir/src/hardware/robot.cpp.o
[ 12%] Building CXX object MakeFiles/hightorque_robot.dir/src/crc/crc8.cpp.o
[ 12%] Building CXX object MakeFiles/hightorque_robot.dir/src/crc/crc16.cpp.o
[ 13%] Linking CXX shared library libhightorque_robot.so
[ 13%] Built target hightorque_robot
Scanning dependencies of target parse_demo
[ 13%] Building CXX object MakeFiles/parse_demo.dir/example/parse_demo.cpp.o
[ 14%] Linking CXX executable parse_demo
[ 14%] Built target parse_demo
Scanning dependencies of target motor_run
[ 15%] Building CXX object MakeFiles/motor_run.dir/example/motor_run.cpp.o
[ 15%] Linking CXX executable motor_run
[ 16%] Built target motor_run
Scanning dependencies of target motor_msg_subscriber
[ 17%] Building CXX object MakeFiles/motor_msg_subscriber.dir/example/motor_msg_subscriber.cpp.o
[ 17%] Linking CXX executable motor_msg_subscriber
[ 17%] Built target motor_msg_subscriber
Scanning dependencies of target hightorque_robot_py
[ 17%] Building CXX object MakeFiles/hightorque_robot_py.dir/cython/robot_wrapper.cpp.o
[ 17%] Building CXX object MakeFiles/hightorque_robot_py.dir/cython/robot_wrapper.cpp.o
```

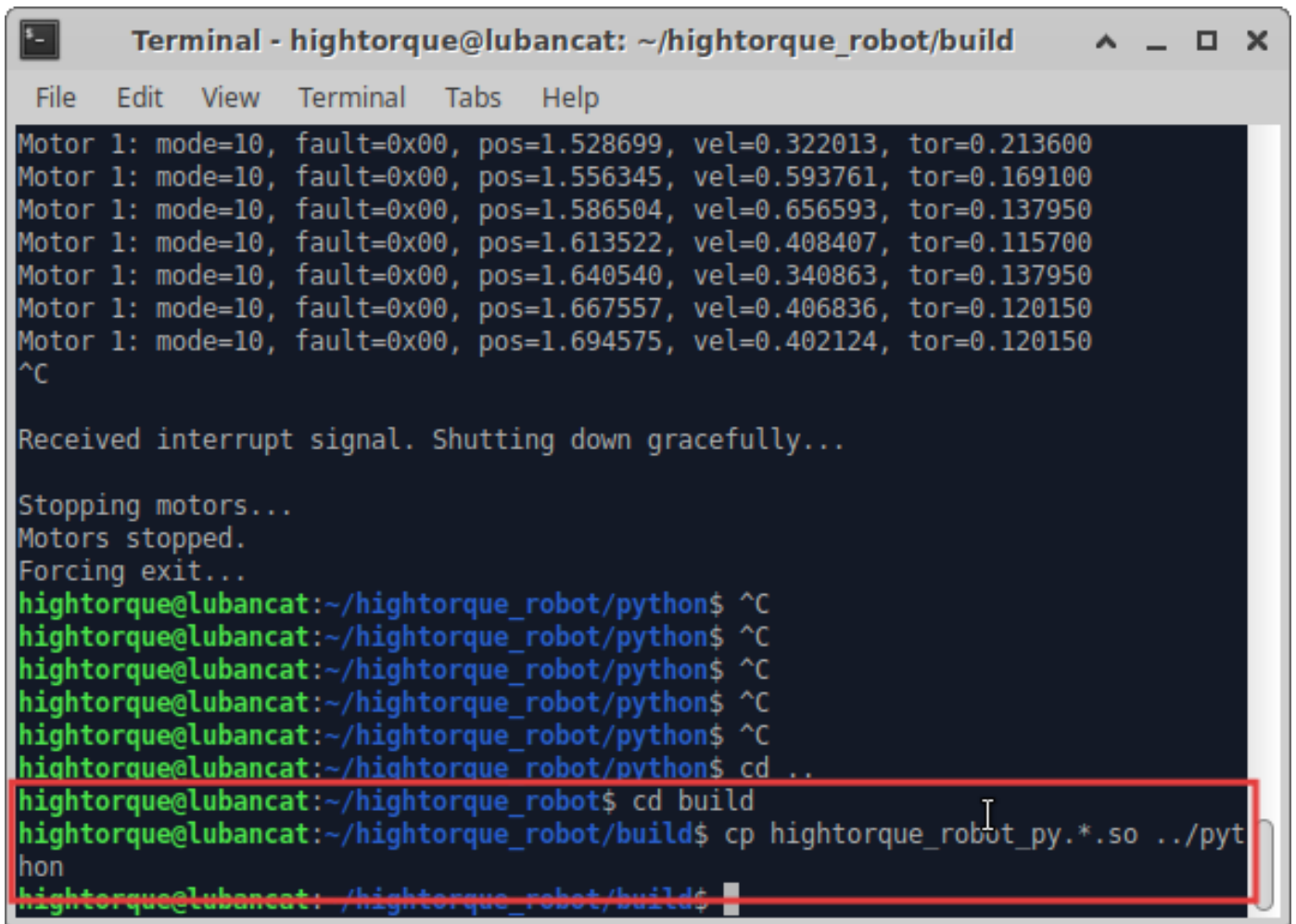
4. 将在build文件中生成的hightorque_robot_py*.so文件移到python

- 使用下面指令进行操作，检查是否把文件移进python文件下。

代码块

```
1 cp hightorque_robot_py*.so ../python
```


- `hightorque_robot_py.*.so` 根据python版本不同此处名称有所不同，例如使用python3.8，生成的是 `cpython-38`，如下图中的 `hightorque_robot_py.cpython-38-aarch64-linux-gnu.so`。

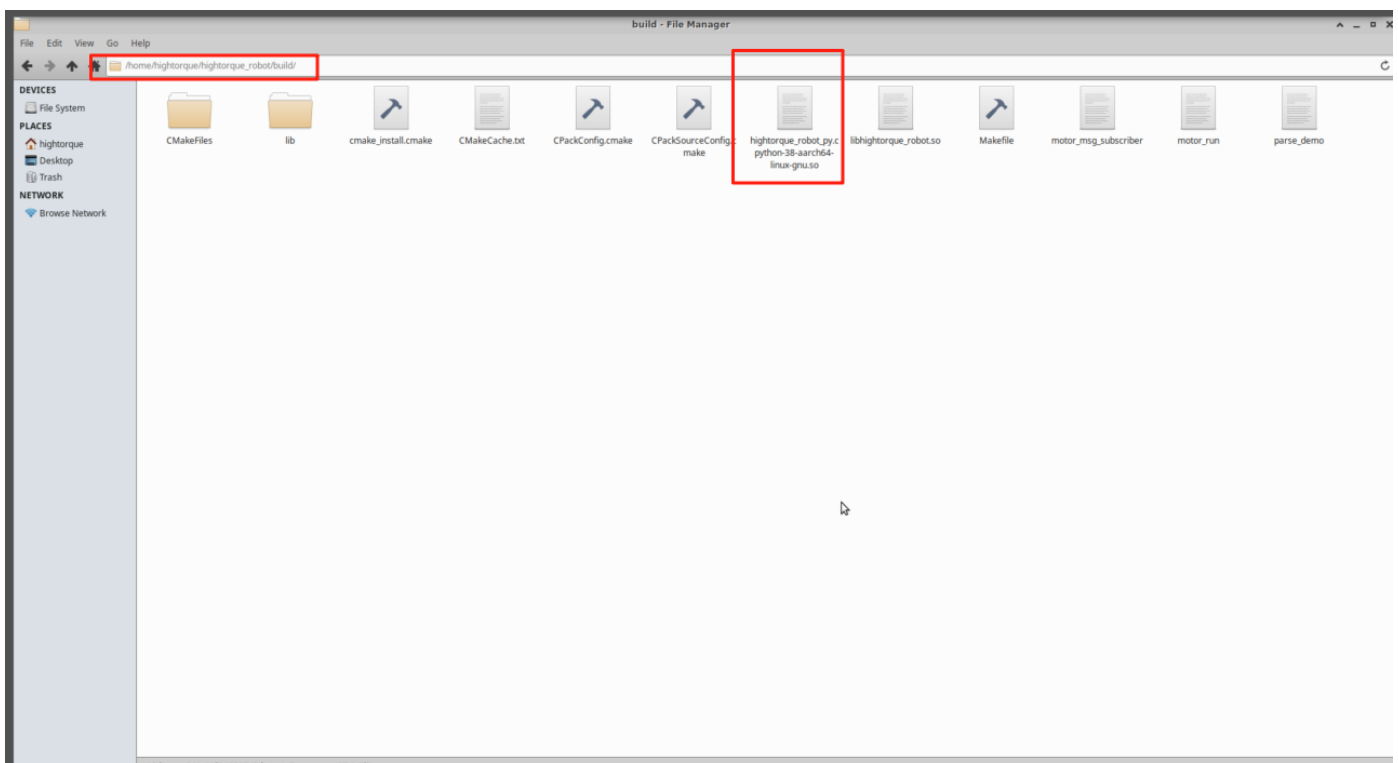


```
Terminal - hightorque@lubancat: ~/hightorque_robot/build
File Edit View Terminal Tabs Help

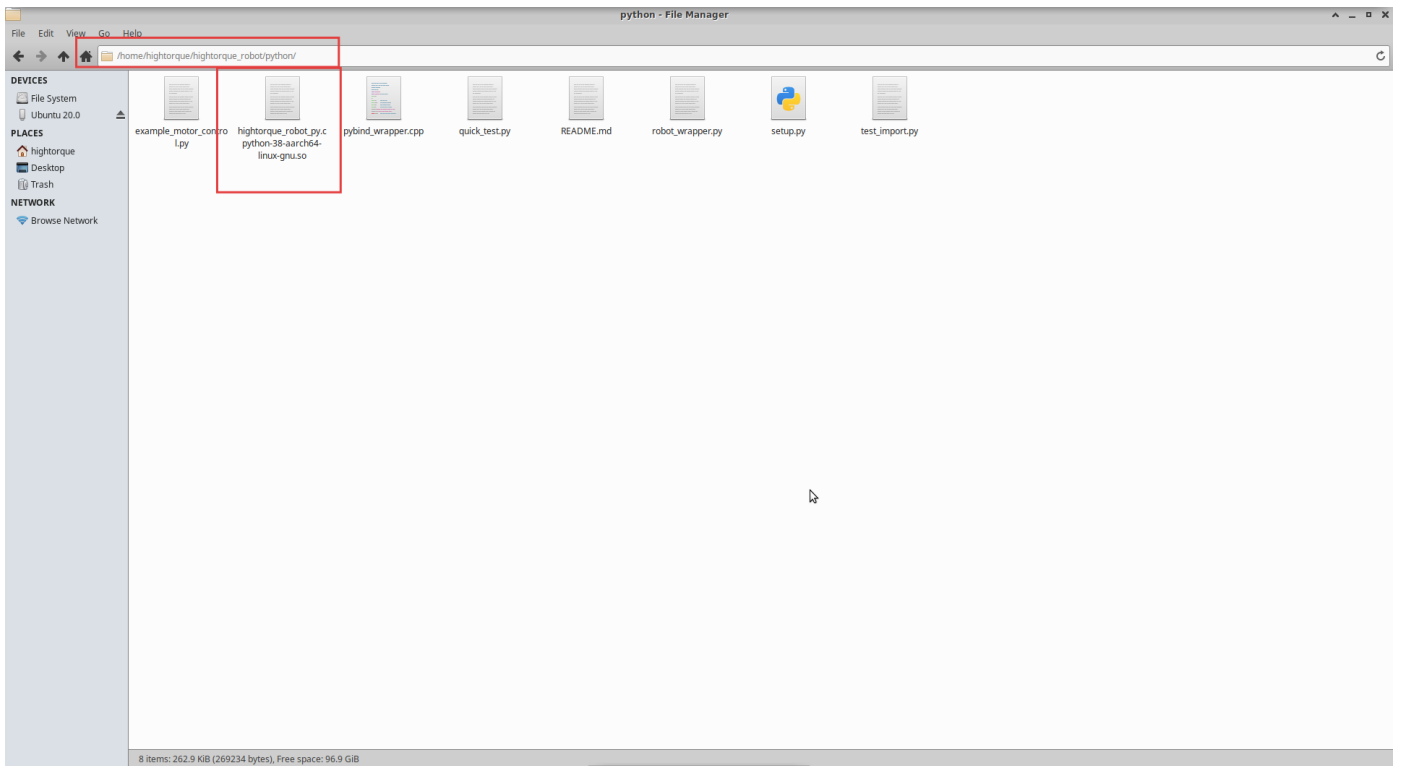
Motor 1: mode=10, fault=0x00, pos=1.528699, vel=0.322013, tor=0.213600
Motor 1: mode=10, fault=0x00, pos=1.556345, vel=0.593761, tor=0.169100
Motor 1: mode=10, fault=0x00, pos=1.586504, vel=0.656593, tor=0.137950
Motor 1: mode=10, fault=0x00, pos=1.613522, vel=0.408407, tor=0.115700
Motor 1: mode=10, fault=0x00, pos=1.640540, vel=0.340863, tor=0.137950
Motor 1: mode=10, fault=0x00, pos=1.667557, vel=0.406836, tor=0.120150
Motor 1: mode=10, fault=0x00, pos=1.694575, vel=0.402124, tor=0.120150
^C

Received interrupt signal. Shutting down gracefully...

Stopping motors...
Motors stopped.
Forcing exit...
hightorque@lubancat:~/hightorque_robot/python$ ^C
hightorque@lubancat:~/hightorque_robot/python$ ^C
hightorque@lubancat:~/hightorque_robot/python$ ^C
hightorque@lubancat:~/hightorque_robot/python$ ^C
hightorque@lubancat:~/hightorque_robot/python$ ^C
hightorque@lubancat:~/hightorque_robot/python$ cd ..
hightorque@lubancat:~/hightorque_robot$ cd build
hightorque@lubancat:~/hightorque_robot/build$ cp hightorque_robot_py.*.so ../python
hightorque@lubancat:~/hightorque_robot/build$
```



build文件下文件信息



python文件下文件信息

5. 进入python文件运行测试程序

- 运行下面指令

代码块

```
1 cd ..  
2 cd python\  
3 python3 example_motor_control.py
```

- 运行结果如下，python脚本运行，执行电机控制程序


```
Terminal - hightorque@lubancat: ~/hightorque_robot/python ^ _ □ ×
File Edit View Terminal Tabs Help
mq_test.cpp.o
[ 97%] Building CXX object lib/lcm/test/cpp/CMakeFiles/test-cpp-memq_test.dir/co
mmon.cpp.o
[ 98%] Linking CXX executable test-cpp-memq_test
[ 98%] Built target test-cpp-memq_test
Scanning dependencies of target test-cpp-client
[ 99%] Building CXX object lib/lcm/test/cpp/CMakeFiles/test-cpp-client.dir/clien
t.cpp.o
[ 99%] Building CXX object lib/lcm/test/cpp/CMakeFiles/test-cpp-client.dir/commo
n.cpp.o
[100%] Linking CXX executable test-cpp-client
[100%] Built target test-cpp-client
hightorque@lubancat:~/hightorque_robot/build$
hightorque@lubancat:~/hightorque_robot/build$ cd ..
hightorque@lubancat:~/hightorque_robot$ cd python/
hightorque@lubancat:~/hightorque_robot/python$ python3 example_motor_control.py
Hightorque Robot Python Interface Demo
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Available motor types:
General: MotorType.mGeneral
60SG_35: MotorType.60SG_35
7256_35: MotorType.m7256_35
5047_36_2: MotorType.5047_36_2
```